

Ultra-compact Reflow-compatible Detachable Optical Connector for Co-Packaged Optics

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Abstract: Ultra-compact detachable optical connector for Co-Packaged Optics is developed. The connector demonstrates low insertion loss ≤ 0.4 dB with small variation within ± 0.05 dB in multiple cycles connections and 260 °C reflow-compatibility.

1. Introduction

Co-packaged Optics (CPO) is highly anticipated as a key enabler to realize higher capacity and lower power consumption of optical communication in next generation data center (DC) [1, 2]. In CPO, photonic integrated circuits (PICs) based on silicon photonic (SiPh) are placed close to application specific integrated circuit (ASIC) and function as optical transceivers. For input and output of the light to and from PIC in the practical use, the optical connectivity device for PIC/optical fiber connection is necessary. To avoid the complexity of optical fiber handling during the packaging process of CPO substrate, it is preferable to adopt a detachable optical connector [3-4] and remove the optical fiber side connector terminal during the packaging process. In addition, since the CPO substrate is packaged by the reflow process, the reflow-compatibility is also required for the optical connector terminal on the CPO substrate side. Furthermore, in recent years, it has been reported that a glass substrate rather than an organic substrate is applied as a substrate for CPO [5]. The glass substrate has an advantage of low warpage and low coefficient of thermal expansion (CTE). It is also possible to form an interface for pick-and-place of the PIC and connection with an external optical fiber by forming a waveguide inside of the glass substrates.

In this paper, we propose an ultra-compact and detachable optical connector for CPO application using a coupling by a collimated beam with magnetic force connection. We report on the insertion loss characteristics and reflow-compatibility. In addition, we demonstrate a concept of high-density multiple-series packaging of the optical connectors on glass substrate for CPO by taking advantage of the compact size of the connector.

2. Design concept of detachable optical connector employing collimated beam and magnetic force connection

In this paper, we develop an optical connector with a 12-channel, 250- μm pitch interface. Figure 1 shows a design concept for detachable waveguide/optical fiber connector. A waveguide in glass substrate and fiber array (FA) are connected with MT connector-like registration parts. The registration part is made of polyphenylene sulfide (PPS) and is manufactured by injection molding with high productivity. Lens arrays (LAs) are precisely positioned with the registration parts with reference to the guide mechanism and fixed, and the waveguide in the glass substrate and the FA are actively aligned and fixed to a rear surface of the LAs respectively. The light emitted from the waveguide and the optical fiber is collimated and focused by a pair of lens arrays and coupled to the core of the opposing optical fiber and the optical waveguide. By expanding the beam diameter by the lens, the tolerance for misalignment between the registration parts is relaxed. In this study, the beam diameter of the collimated beam is set to 60 μm (taken at $1/e^2$), and in this case, the misalignment in which the coupling loss deteriorates by 0.2 dB is enlarged to about 6.5 μm from about 1 μm in a connection of standard single mode fibers.

In addition, by employing a non-contact connection instead of the general physical contact (PC) in a multi-fiber connector, a strong spring mechanism for PC connection and a robust connector housing can be eliminated, the connector terminal can be downsized. In the proposed structure, magnets are fixed to both sides of the registration part, and mating forces are applied to connector terminals. When the opposing connector terminals come close to each other to a few millimeters, the connector pairs can be connected automatically to each other by the magnetic force. Hence, the workability of the edge connection in the limited space is secured. The force required to remove the magnetic connection is approximately 2.2 N, and the magnetic connection is considered to have a practical strength.

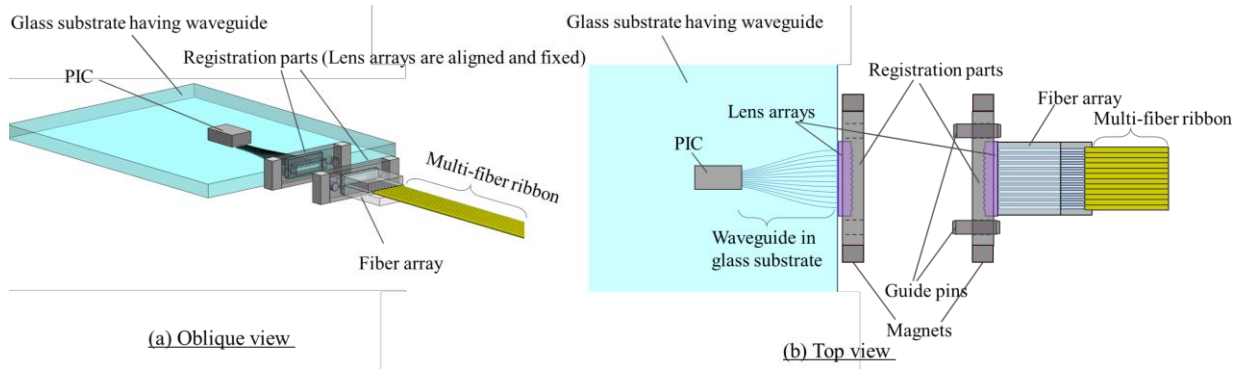


Fig. 1. Design concept for detachable Waveguide/ Optical fiber connector.

3. Evaluation of insertion loss and reflow-compatibility

First, the loss characteristics of the optical connector are evaluated in connections with FAs as shown in Fig. 2 to confirm the principle of the proposed optical connector. The optical fiber constituting FAs is compliant with ITU-T G. 657, and its mode field diameter (MFD) is $8.6 \mu\text{m}$ at 1310 nm . The size of the one connector terminal is 2.5 mm in height, 8.4 mm in width, and 5.7 mm in length, and the pitch of the guide hole and hole diameter are 5.2 mm and 0.7 mm , respectively. Figure 3 shows the variation of insertion loss at 1310 nm at 10 cycles connections measure for two connection pairs (Sample A and B). Low insertion loss $\leq 0.4 \text{ dB}$ with small variation within $\pm 0.05 \text{ dB}$ is demonstrated in 10 cycles connections. Accordingly, the proposed optical connector structure is conducive to superior and stable loss characteristics.

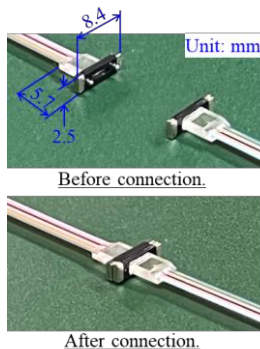


Fig. 2. Detachable optical connector for FA/FA connection.

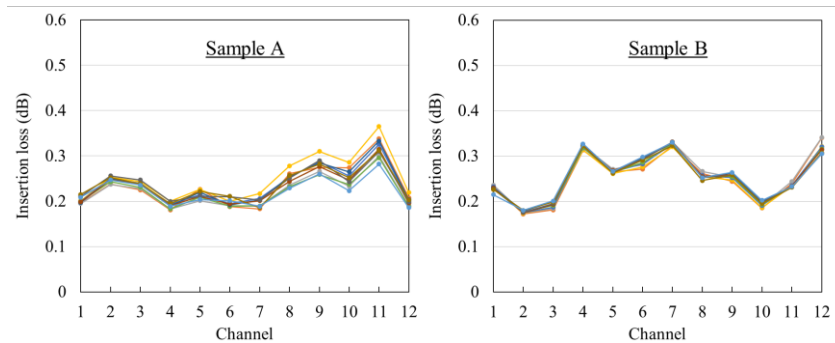


Fig. 3. Insertion Loss variation at 1310 nm in 10 cycles connections for FA/FA connections (for two samples).

Next, the reflow-compatibility of the developed optical connector is evaluated in case of the connections with planar lightwave circuit (PLC) having 12-channel waveguide instead of CPO glass substrate. As shown in Fig. 4, two FAs are connected to both ends of PLC with the developed optical connectors. The MFD of the PLC is designed to $8.6 \mu\text{m}$ in order to match that of the optical fiber. During the reflow, the FA-side optical connectors are removed, and only the PLC and two optical connector terminals fixed to both ends of PLC are exposed to the reflow environment. In this experiment, the temperature profile with peak temperature of 260°C and 60 seconds duration is applied as a condition close to reflow.

Figure 5 shows the insertion loss before and after the reflow for one sample in five cycles connections at both PLC end faces, and plots and bars correspond to the average and maximum/minimum insertions loss in five cycles connections. The insertion loss includes loss of two FA/PLC connections and waveguide loss. The insertion loss before reflow is less than 1 dB , and the loss per one PLC/fiber connection is estimated to less than 0.5 dB . The change of the average losses before and after reflow is less than 0.14 dB and verified small. The loss variations for 12 channels in five cycles connections are within $\pm 0.05 \text{ dB}$ both for before/after reflow and very stable. From these results, it is confirmed that the developed connector has the reflow-compatibility.

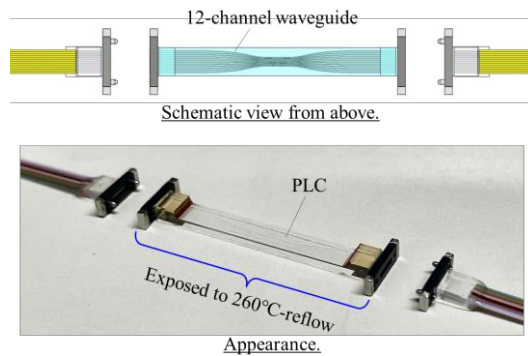


Fig. 4. Configuration of PLC/FAs connection for reflow-compatibility evaluation.

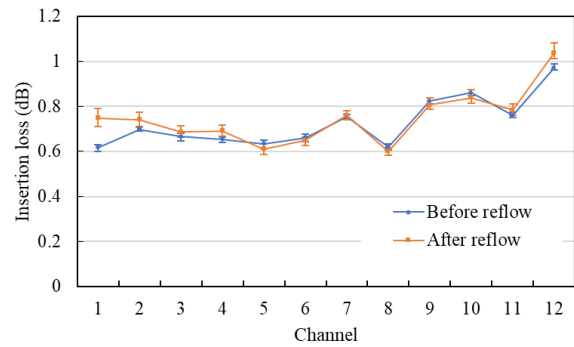


Fig. 5. Insertion loss at 1310nm for PLC/FAs connection before and after reflow in 5 cycles connections at both end faces.

4. Multiple series connection with glass substrate

We fabricate a mock-up of multiple series connection with glass substrate in CPO. Figure 6 shows the mock-up of four series connections of FAs to the glass substrate. The size of glass substrate is 50 mm x 50 mm x 2.2 mm thick, and the connection period of optical connectors is 10 mm. In the case of 12-channel optical connector, the channel density is as high as 1.2 channel/mm. Even in such a spatially dense connection, the developed connector can be connected to the target optical connector without the improper connection with the self-alignment mechanism of magnetic force connection.

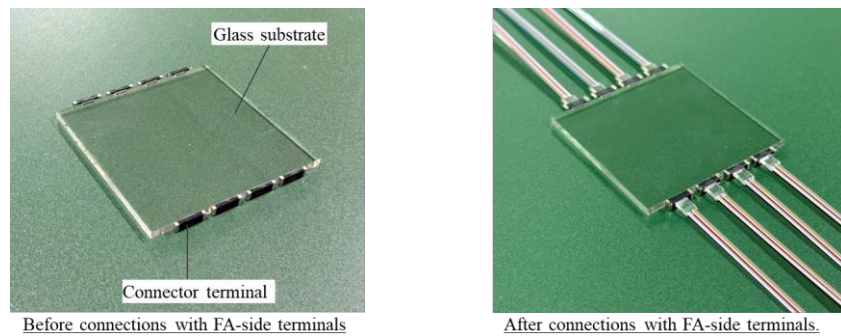


Fig. 6. Concept demonstration of four-series connection of FAs to the CPO glass substrate

5. Conclusions

Ultra-compact detachable optical connector utilizing collimated beam and magnetic force was developed for optical waveguide/optical fiber connection in CPO. Low insertions loss ≤ 0.4 dB and small loss variation within ± 0.05 was demonstrated in 10 cycles connections of FAs. This connection structure is suitable to an ultra-compact optical connector in CPO. In addition, the compatibility to 260 °C reflow was demonstrated in the connection with PLC. Moreover, we mocked up four series connection of FAs to glass substrate with high packaging density of 1.2 channel/mm.

6. Acknowledgement

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7. References

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